KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

TOPIC: MICROCONTROLLER-BASED WATER TANK LEVEL MONITORING, FILLING AND TURBINE CONTROL SYSTEM

GROUP 18

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ABSTRACTS

This project presents the design and implementation of a microcontroller-based water tank level monitoring, filling, and turbine control system. The system aims to automate water management by continuously monitoring the water level in a tank and controlling the filling process using a pump. Additionally, a turbine mechanism is incorporated to simulate energy generation from water flow. The system uses sensors to detect water levels and a microcontroller to process the data and control outputs such as the pump and turbine. Simulation results demonstrate that the system effectively prevents water overflow, avoids dry running of pumps, and improves overall efficiency. This project highlights the importance of automation in resource management and provides a foundation for future smart water systems.

CHAPTER ONE

1. INTRODUCTION

Water is one of the most essential resources for human survival, agriculture, and industrial activities. Efficient management of water resources is therefore critical, especially in regions where water supply is inconsistent or limited. In many homes, institutions, and industries, water is stored in overhead or underground tanks and is often managed manually. This manual approach to water tank management is inefficient and can lead to several problems such as overflow, water wastage, pump damage due to dry running, and unnecessary energy consumption.

With the advancement of embedded systems and automation technologies, microcontrollers have become widely used in solving real-world problems due to their reliability, low cost, and ease of integration with sensors and actuators. A microcontroller-based system can continuously monitor environmental conditions and make intelligent decisions without human intervention. This makes it an ideal solution for automating water tank operations.

In this project, a microcontroller-based water tank level monitoring, filling, and turbine control system is developed to address these challenges. The system uses water level sensors to detect the amount of water in a tank and automatically controls a pump to maintain an optimal water level. When the water level drops below a predefined threshold, the system activates the pump to refill the tank. Once the tank reaches its maximum level, the pump is automatically turned off to prevent overflow.

In addition to monitoring and filling, this project introduces a turbine control mechanism. The turbine, simulated using a DC motor, is activated when water flows through the system. This demonstrates the concept of energy generation from water movement, which can be further developed into real-world applications such as micro-hydropower systems.

The integration of monitoring, control, and energy simulation into a single system makes this project both innovative and practical. It not only improves water management efficiency but also reduces human effort and operational risks. Furthermore, the system can be expanded in the future by incorporating Internet of Things (IoT) technologies, enabling remote monitoring and control via mobile or web applications.

Overall, this project demonstrates how microcontroller-based automation can be effectively applied to solve everyday problems in a smart and efficient manner.

CHAPTER 2

2. PROBLEM STATEMENT

Many water tank systems rely on manual operation, which can lead to:

- Water overflow due to lack of monitoring
- Dry running of pumps causing damage
- Inefficient water usage
- Lack of integration with energy-generating systems like turbines

This project seeks to design a system that automatically:

- Monitors water levels
- Controls pump operation (filling)
- Activates a turbine based on water flow

2.1 OBJECTIVES

- To design a microcontroller-based water level monitoring system
- To automate the filling of the tank using sensors
- To integrate a turbine control mechanism
- To simulate and demonstrate the system functionality

CHAPTER 3

3. SYSTEM COMPONENTS

4.1 Hardware Components

- Arduino Uno
- LED
- Breadboard
- Ultrasonic sensor
- Buzzer
- Resistor
- Jumpers
- HR Motor Driver

3.2 Software Tools

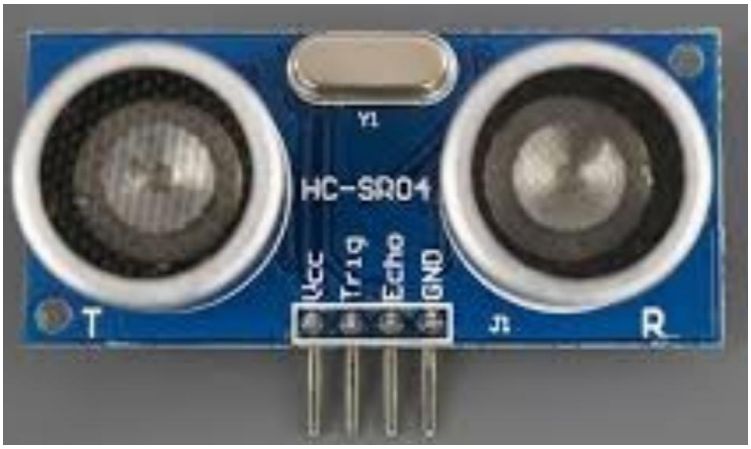
- Arduino IDE
- Simulation software (Tinkercad)

3.3 METHODOLOGY

- The system was designed as an automated water tank monitoring and control system using a microcontroller, where all components (motor, buzzer, and LED) work together to ensure efficient operation without human intervention.
- An ultrasonic sensor was used to measure the water level in the tank by sending ultrasonic waves to the water surface and receiving the reflected signal. The time taken for the echo to return was used to calculate the distance between the sensor and the water, allowing the system to determine how full the tank is.
- The microcontroller (Arduino Uno) was programmed to continuously read the distance values from the ultrasonic sensor and compare them with predefined threshold levels. These thresholds were used to classify the water level as low, rising, or near full.
- Based on the processed sensor data, the microcontroller controls the system outputs automatically using conditional logic. This ensures that the system responds immediately to any change in water level.
- A DC motor was incorporated into the system to simulate the operation of a water pump. When the water level is low, the microcontroller activates the motor to represent the pumping of water into the tank.

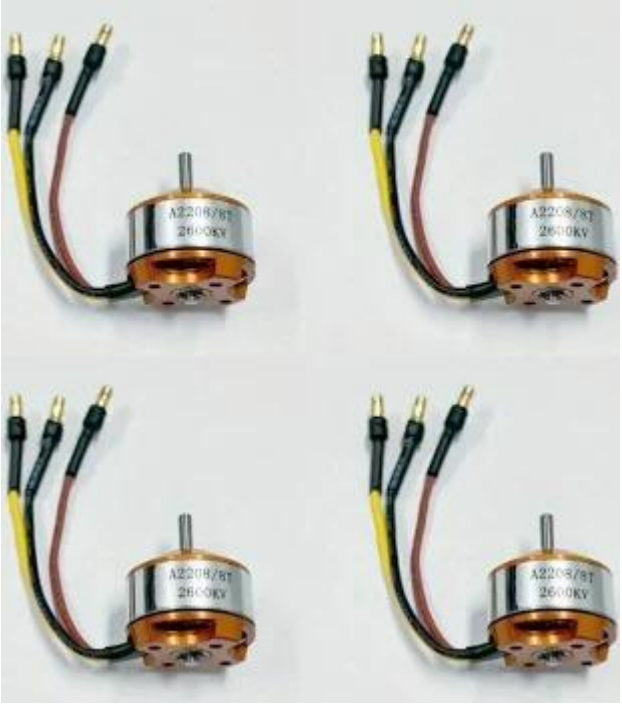
- As the water level continues to rise, the motor remains in operation until the water reaches a near-full condition, at which point the microcontroller turns the motor OFF to prevent overflow and unnecessary energy usage.
- An alert system consisting of a buzzer and a red LED was implemented to notify the user when the water level is close to the top of the tank. At this point, the microcontroller activates both the buzzer and LED to provide audible and visual warnings.

LED



ULTRASONIC SENSOR

ARDUINO UNO



MOTOR

3.4 CODE OVERVIEW

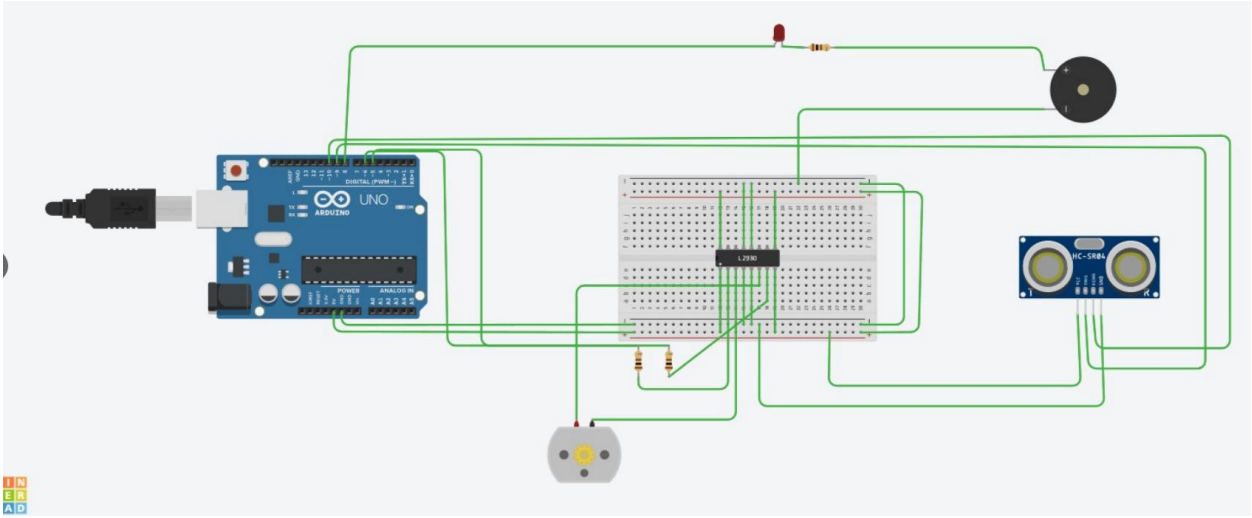
The system was programmed using C++ in the Arduino IDE, which serves as the control logic for the entire water tank monitoring system. The code is designed to continuously read data from the ultrasonic sensor and control the motor, buzzer, and LED based on the detected water level.

The program begins with the declaration and initialization of variables and pin configurations. The ultrasonic sensor pins (trigger and echo) are defined, while the motor, buzzer, and LED pins are set as output devices using C++ syntax. A function is implemented to calculate the distance between the ultrasonic sensor and the water surface. This is achieved by sending a pulse through the trigger pin and measuring the duration of the returned echo signal. The duration is then converted into distance. This allows the system to monitor water levels in real time.

Conditional statements (if, else if, else) are used to compare the measured distance with predefined threshold values. These conditions determine the state of the system (low level or near full level). When the measured distance indicates a low water level, the code activates the motor to simulate the pumping process, while keeping the buzzer and LED turned OFF. As the water level rises and reaches a near-full condition (i.e., the distance becomes small), the code turns OFF the motor and activates the buzzer and red LED to alert the user.

CHAPTER 4

5.1 SIMULATION/DEMONSTRATION



5.2 SIMULATION PROCESS

The system was implemented using Tinkercad to model the operation of the water tank control system.

All required components, including the microcontroller, ultrasonic sensor, motor, buzzer, and LED, were selected and assembled in the simulation environment according to the circuit design.

Proper connections were made between the components, ensuring that input devices (ultrasonic sensor) and output devices (motor, buzzer, LED) were correctly interfaced with the microcontroller pins.

The C++ program was written in the Arduino IDE and then uploaded into the simulated microcontroller to control the system.

5.3 OBSERVED OUTPUT

- Pump activates automatically when tank is empty
- Pump stops when tank is full

- Ultrasonic sensor detects water level

CONCLUSION

The project successfully demonstrates an automated water tank monitoring and control system using a microcontroller. It improves efficiency, reduces water wastage, and introduces a basic turbine control mechanism. This system can be applied in homes, industries, and agricultural settings.

CHALLENGES ENCOUNTERED

- Limited components
- High starting current of the dc motor

RECOMMENDATIONS/ FUTURE IMPROVEMENT

- Improve turbine efficiency for real power generation
- Use more accurate sensors